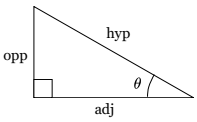
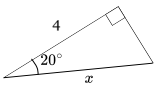
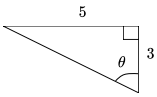
Trigonometry

Chapter 9 · Original slide 4 · RECAP: Right-Angled Trigonometry





Find x.



Find θ.

You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$

But what is the *conceptual* definition of $\sin$?

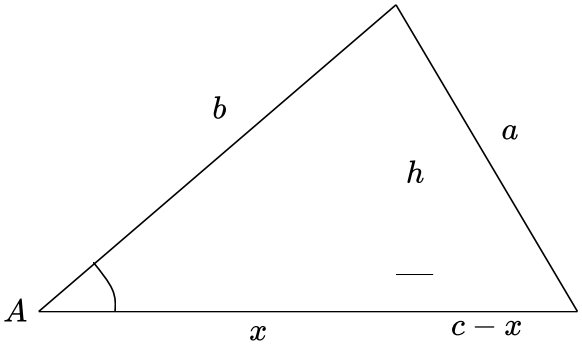
sin is a function which inputs an angle and gives the ratio between the opposite and hypotenuse.

Remember that a ratio just means the 'relative size' between quantities (in this case lengths). For this reason, sin/cos/tan are known as "trigonometric ratios".

reflections: You may have been taught "use tan^{-1} whenever you're finding an angle", and therefore write the second line directly. It is preferable to always write the first line, then see the problem as a 'changing the subject' one. We need to remove the tan on front of the θ, so apply tan^{-1} to each side of the equation to 'cancel out' the tan on the LHS.

Tip: You can swap the thing you're dividing by and the result. e.g. $\frac{8}{2} = 4 \rightarrow \frac{8}{4} = 2$.

Chapter 9 · Original slide 5 · Proof of Cosine Rule



We want to use Pythagoras, so split c into two so that we get two right-angled triangles.

By Pythagoras: $h^2 + x^2 = b^2$ and
$h^2 + (c - x)^2 = a^2$

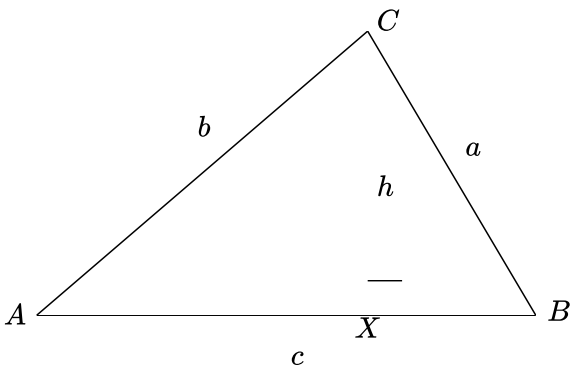
Subtracting to eliminate h:

$x^2 - (c - x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$

But $x = b\cos A \therefore$

$a^2 = b^2 + c^2 - 2bc\cos A$

Chapter 9 · Original slide 6 · Proof of Sine Rule



The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.

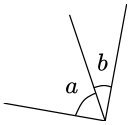
Using basic trigonometry:

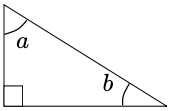
$h = a\sin B$ and $h = b\sin A$

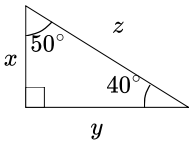
$\therefore a\sin B = b\sin A$

$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

Chapter 9 · Original slide 7 · Just for your interest…







Why does “cosine” contains the word “sine”?

Complementary Angles

add to 90°

Therefore these angles are complementary.

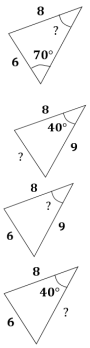
i.e. The cosine of an angle is the sine of the complementary angle.

Hence cosine = COMPLEMENTARY SINE

$\therefore \mathbf{cos\,(50) = sin\,(40)}$

Trigonometry

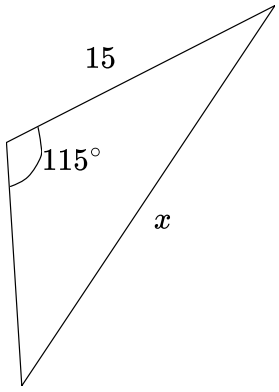
Chapter 9 · Original slide 8 · OVERVIEW: Finding missing sides and angles



When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	
#2 Two sides known and a missing side opposite a known angle	Remaining side	
#3 All three sides	An angle	
#4 Two sides known and a missing side not opposite known angle	Remaining side	

Chapter 9 · Original slide 9 · Cosine Rule



We use the cosine rule whenever we have three sides (and an angle) involved.

b

Cosine Rule:

$a^2 = b^2 + c^2 - 2bc\cos A$

Proof at end of PowerPoint.

a A

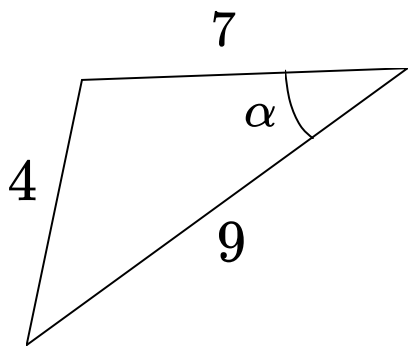
The only angle in formula is A, so label angle in diagram A, label opposite side a, and so on (b and c can go either way).

c

12

Trigonometry

Chapter 9 · Original slide 10 · Dealing with Missing Angles



You have	You want	Use
All three sides	An angle	Cosine rule

$\mathbf{a^2 = b^2 + c^2 - 2bc\cos A}$

Tip: The brackets are not needed, but students who forget about BIDMAS see $4^2 = \left(7^2 + 9^2 - 2 \times 7 \times 9\right)\cos\alpha$ and hence incorrectly simplify to $16 = 4\cos\alpha$

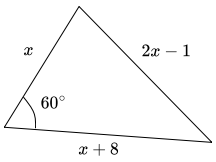
Label sides then substitute into formula.

Simplify each bit of formula.

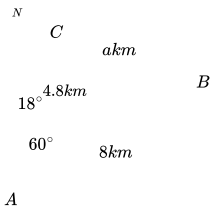
Rearrange "swapping" thing you're subtracting and result

Textbook Note: The textbook presents the rearrangement of the cosine rule: $\cos A = \frac{b^2+c^2-a^2}{2bc}$ to find missing angles. It is not advisable to memorise this

Chapter 9 · Original slide 11 · Harder Ones

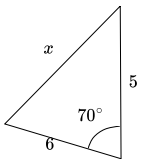


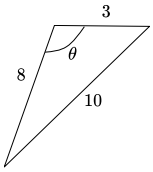
Determine the value of x.

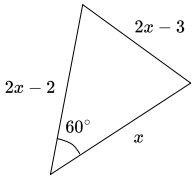


[From textbook] Coastguard station B is 8 km, on a bearing of $060°$, from coastguard station A. A ship C is 4.8 km on a bearing of $018°$, away from A. Calculate how far C is from B.

Chapter 9 · Original slide 12 · Test Your Understanding





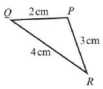


1 3 2

Note: You will get an obtuse angle whenever you inverse cos of a negative value.

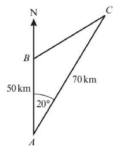
Chapter 9 · Original slide 13 · Exercise 9A – Y1 pages 177-179

8 Show that $\cos P = -\frac{1}{4}$.



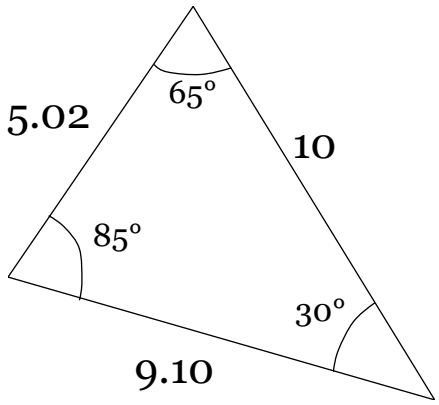
11 The lengths of the sides of a triangle are in the ratio $2 : 3 : 4$. Calculate the size of the largest angle.

Chapter 9 · Original slide 14 · Exercise 9A – Y1 pages 177-179



17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is $020°$.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)

Chapter 9 · Original slide 16 · The Sine Rule



C B a

! Sine Rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

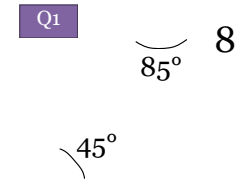
c b A

For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

Chapter 9 · Original slide 17 · Examples

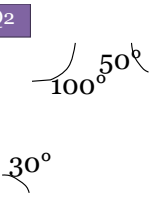
Q1



$\frac{x}{\sin 85} = \frac{8}{\sin 45}$

$x = \frac{8 \sin 85}{\sin 45} = 11.27$

Q2



$\frac{x}{\sin 100} = \frac{8}{\sin 30}$

$x = \frac{8 \sin 100}{\sin 30} = 15.76$

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

Chapter 9 · Original slide 18 · Examples

Q3



$\frac{\sin \theta}{8} = \frac{\sin 126^\circ}{10}$

$\sin \theta = \frac{8 \sin 126}{10}$

$\theta = \sin^{-1}\left(\frac{8 \sin 126}{10}\right)$

$= 40.33^\circ$

Q4



When you have a missing angle, it's better to reciprocate to get:

$\frac{\sin A}{a} = \frac{\sin B}{b}$

i.e. in general put the missing value in the numerator.

$\frac{\sin \theta}{5} = \frac{\sin 85}{6}$

$\sin \theta = \frac{5 \sin 85}{6}$

$\theta = \sin^{-1}\left(\frac{5 \sin 85}{6}\right)$

$= 56.11^\circ$

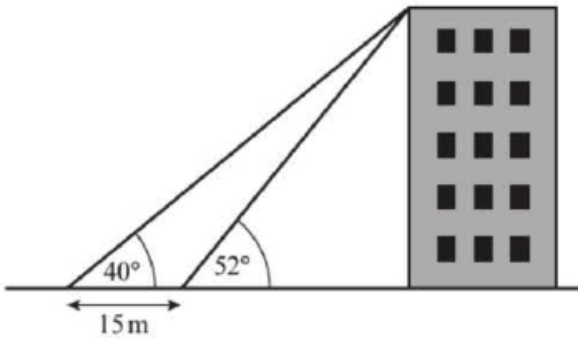
Chapter 9 · Original slide 19 · Exercise 9B – Y1 pages 181-183

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B.
Work out the distance of town C from:
a town A $btownB$

Chapter 9 · Original slide 20 · Exercise 9B – Y1 pages 181-183

11 In $\triangle ABC$, $x = 4(\sqrt{2} - 1)$
$AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and $\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

Chapter 9 · Original slide 21 · Exercise 9B – Y1 pages 181-183

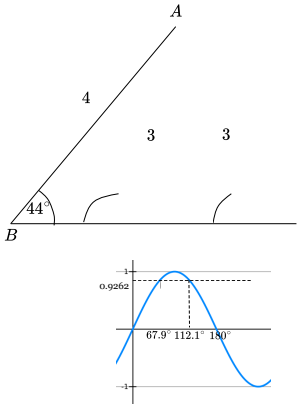


12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.

a Use this information to determine the height of the building. (4 marks)

b State one assumption made by the surveyor in using this mathematical model. (1 mark)

Chapter 9 · Original slide 23 · The 'Ambiguous Case'



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44°$. What are the possible values of $\angle ACB$?

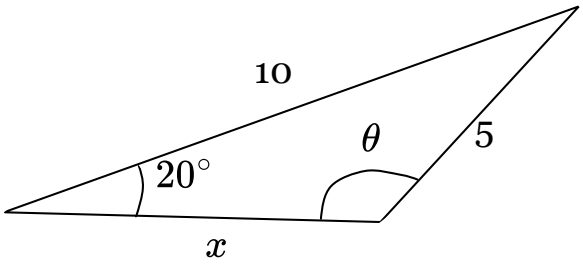
Your calculator will give the acute angle of $67.9°$ (i.e. C_2). But if we look at a graph of sin, we can see there's actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

! The sine rule produces two possible solutions for a missing angle:

$\sin\theta = \sin(180° - \theta)$

Whether we use the acute or obtuse angle depends on context.

Chapter 9 · Original slide 24 · Test Your Understanding



Given that the angle θ is obtuse, determine θ and hence determine the length of x.

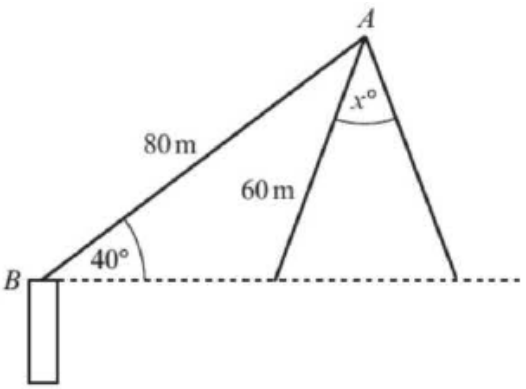
Chapter 9 · Original slide 25 · Exercise 9C – Y1 pages 184-185

4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

Chapter 9 · Original slide 26 · Exercise 9C – Y1 pages 184-185

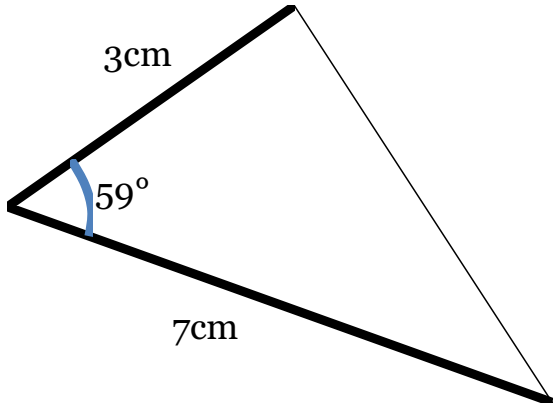
5 Two triangles ABC are such that $AB = 4.5$ cm, $BC = 6.8$ cm and $\angle ACB = 30°$. Work out the value of the largest angle in each of the triangles.

Chapter 9 · Original slide 27 · Exercise 9C – Y1 pages 184-185



6 a A crane arm AB of length 80 m is anchored at point B at an angle of $40°$ to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)

Chapter 9 · Original slide 28 · Area of Non Right-Angled Triangles

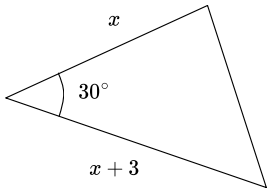


Area = $\frac{1}{2}ab\sin(C)$

where C is the angle between two sides a and b.

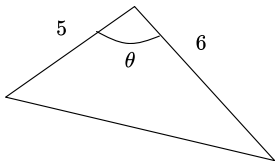
Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides.

Chapter 9 · Original slide 29 · Test Your Understanding



The area of this triangle is 10.

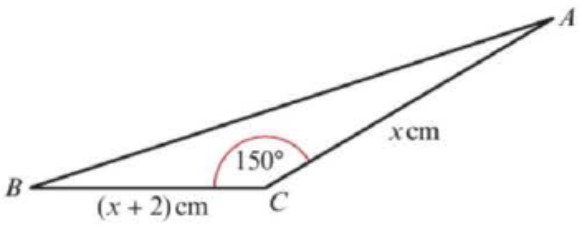
Determine x.



The area of this triangle is also 10.

If θ is obtuse, determine θ.

Chapter 9 · Original slide 30 · Exercise 9D – Y1 pages 186-187



4 In triangle ABC, $BC = (x + 2)$ cm, $AC = x$ cm and $\angle BCA = 150°$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.

Chapter 9 · Original slide 31 · Exercise 9D – Y1 pages 186-187

5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30°$. The area of the triangle is A cm^2.

a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

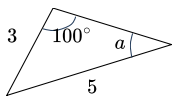
Chapter 9 · Original slide 32 · Exercise 9D – Y1 pages 186-187

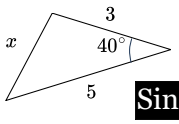
6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle BAC = 150°$. Given that the area of the triangle is $3\frac{3}{4}$ cm^2

a Show that x satisfies the equation $x^2 + 5x - 15 = 0$.

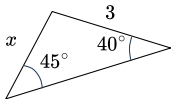
b Calculate the value of x, giving your answer to 3 significant figures.

Chapter 9 · Original slide 33 · Sin or cosine rule?

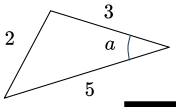




Sine



Sine Cosine



Sine

Recall that whenever we have two “side-angle pairs” involved, use sine rule. If there’s 3 sides involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.

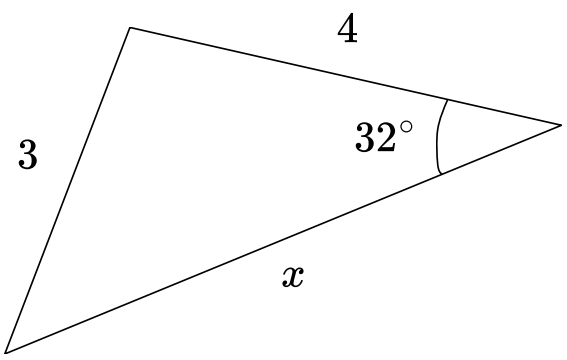
Sine

Cosine

Cosine

Cosine

Chapter 9 · Original slide 34 · Using sine rule twice



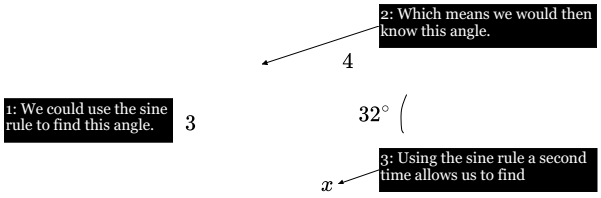
You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

Given there is just one angle involved, you might attempt to use the cosine rule:

This is a quadratic equation!

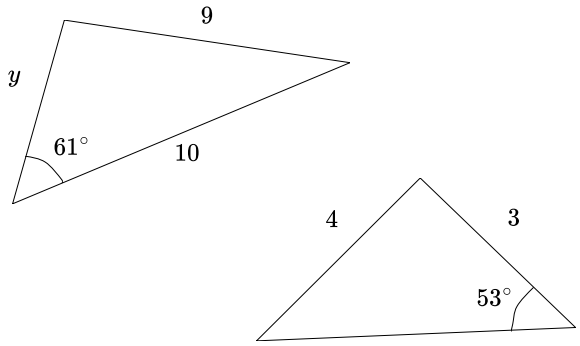
It's possible to solve this using the quadratic formula (using $\mathbf{a=1, b=-8\cos 32, c=7}$). However, this is a bit fiddly and not the primary method expected in the exam...

Chapter 9 · Original slide 35 · Using sine rule twice

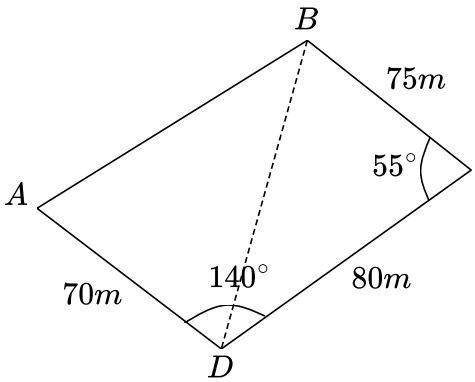


You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

Chapter 9 · Original slide 36 · Test Your Understanding



Chapter 9 · Original slide 37 · Problem Solving With Sine/Cosine Rule



[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$.

$CD = 80m$, angle $BCD = 55°$ and angle $ADC = 140°$.

In order that the masts do not interfere with each other, they must be at least 70m apart.

Given that A is the minimum distance from D, find:

a) The distance A is from B

b) The angle BAD

c) The area enclosed by the four masts.

Chapter 9 · Original slide 38 · Exercise 9E – Y1 page 189-191

12 The perimeter of $\triangle ABC = 15$ cm. Given that $AB = 7$ cm and $\angle BAC = 60°$, find the lengths of AC and BC and the area of the triangle.

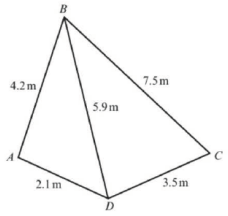
Chapter 9 · Original slide 39 · Exercise 9E – Y1 page 189-191

13 In the triangle ABC, $AB = 14$ cm, $BC = 12$ cm and $CA = 15$ cm.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

Trigonometry

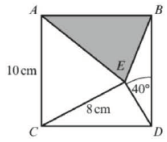
Chapter 9 · Original slide 40 · Exercise 9E – Y1 page 189-191

14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. (4 ma...
b Find the total area of the flowe
(3 ma...
c Find the length of the diagonal AC



Chapter 9 · Original slide 41 · Exercise 9E – Y1 page 189-191

15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)



Chapter 9 · Original slide 44 · Sin Graph

90 180 270 360

What does it look like?

Key Features:

- Repeats every 360°.
- Range: $-1 < \sin(x) < 1$

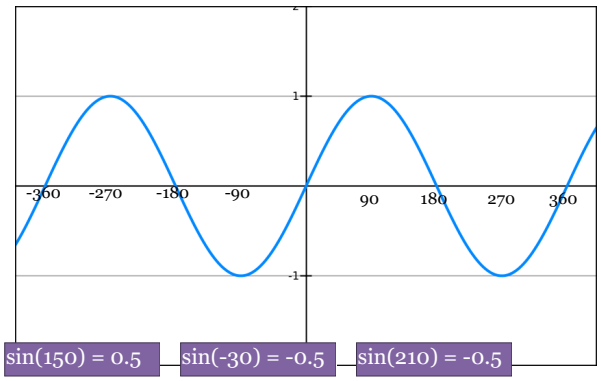
-360

-270

-180

-90

Chapter 9 · Original slide 45 · Sin Graph



What do the following graphs look like?

Suppose we know that sin(30) = 0.5. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 46 · Cos Graph

90 180 270 360

What do the following graphs look like?

Key Features:

- Repeats every 360°.
- Range: $-1 < \sin(x) < 1$

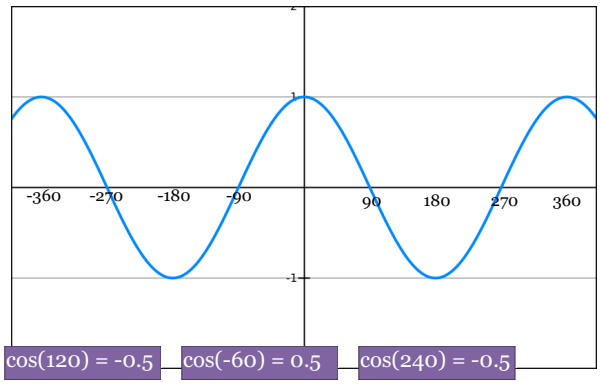
-360

-270

-180

-90

Chapter 9 · Original slide 47 · Cos Graph



What does it look like?

Suppose we know that cos(60) = 0.5. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 48 · Tan Graph

90 180 270 360

What does it look like?

-360

-270

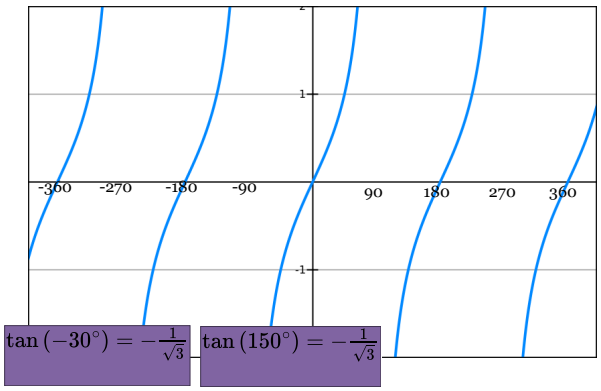
-180

-90

Key Features:

- Repeats every $\mathbf{180}^\circ$.
- Roots: $0, 180^\circ, -180^\circ, \ldots$
- Range: $\tan(x) \in \mathbb{R}$ (i.e. no min/max value!)
- Asymptotes: $x = \pm 90^\circ, \pm 270^\circ, \ldots$

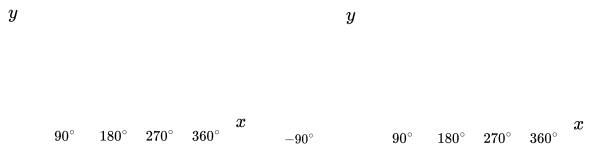
Chapter 9 · Original slide 49 · Tan Graph



What does it look like?

Suppose we know that $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 50 · Transforming Trigonometric Graphs

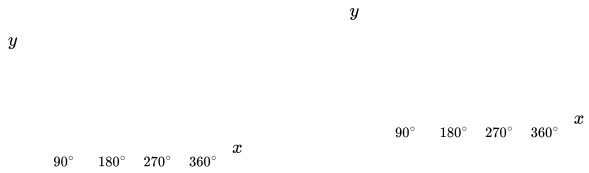


Sketch $y = 4\sin x,\ 0 \leq x \leq 360^\circ$

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs ‘inside’ the function (i.e. input modified) or ‘outside’ the function (i.e. output modified).

Sketch $y = \cos(x + 45^\circ),\ 0 \leq x \leq 360^\circ$

Chapter 9 · Original slide 51 · Transforming Trigonometric Graphs



Sketch $y = \sin\left(\frac{x}{2}\right),\ 0 \leq x \leq 360^\circ$

Sketch $y = -\tan x,\ 0 \leq x \leq 360^\circ$

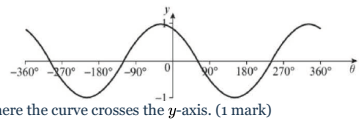
Chapter 9 · Original slide 52 · Exercise 9F – Y1 Page 194

1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

Chapter 9 · Original slide 53 · Exercise 9G – Y1 Pages 197-198

The graph shows the curve $y = \cos(x + 30°)$, $-360° \leqslant x \leqslant 360°$.

a Write down the coordinates of the points where the curve crosses the x-axis. **(2 marks)**

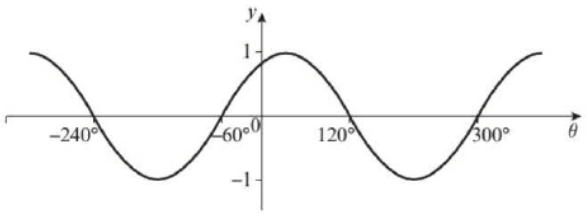


b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

7 The graph shows the curve $y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)

b $\cos 30^\circ = \dfrac{\sqrt{3}}{2}$; $\left(0, \dfrac{\sqrt{3}}{2}\right)$

Chapter 9 · Original slide 54 · Exercise 9G – Y1 Pages 197-198



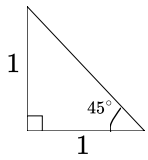
8 The graph shows the curve with equation $y = \sin(x + k)$, $-360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)

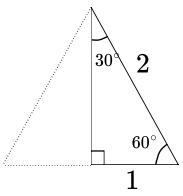
Chapter 9 · Original slide 55 · Exercise 9G – Y1 Pages 197-198

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

Trigonometry

Chapter 10 · Original slide 3 · Trigonometric identities and equations





You will frequently encounter angles of 30°, 60°, 45° in geometric problems. Why?

Although you will always have a calculator, you need to know how to derive these.

All you need to remember:

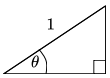
! Draw half a unit square and half an equilateral triangle of side 2.

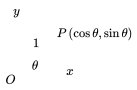
$\sin(30^\circ) =$
$\cos(30^\circ) =$
$\tan(30^\circ) =$
$\sin(60^\circ) =$
$\cos(60^\circ) =$
$\tan(60^\circ) =$

$\sin(45^\circ) =$
$\cos(45^\circ) =$
$\tan(45^\circ) =$

Chapter 10 · Original slide 4 · The Unit Circle and Trigonometry

For values of θ in the range $0 < \theta < 90^\circ$, you know that $\sin\theta$ and $\cos\theta$ are lengths on a right-angled triangle:





But how do we get the rest of the graph for $\sin$, $\cos$ and $\tan$ when $90^\circ \leq \theta \leq 360^\circ$?

! The point P on a unit circle, such that OP makes an angle θ with the positive x-axis, has coordinates $(\cos\theta, \sin\theta)$.
OPhas gradient $\tan\theta$.

And what would be the gradient of the bold line?

Angles are always measured anticlockwise.
(Further Mathematicians will encounter the same when they get to Complex Numbers)

We can consider the coordinate $(\cos\theta, \sin\theta)$ as θ increases from 0 to 360°...

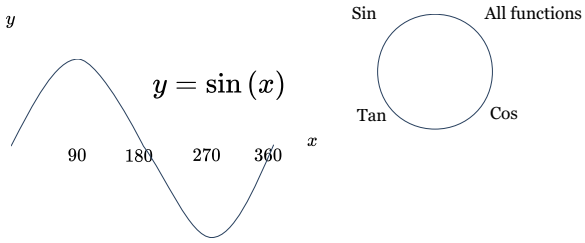
Chapter 10 · Original slide 5 · Mini-Exercise

	$\cos\theta$	$\sin\theta$	$\tan\theta$		$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 0$	1	0	0	$\theta = 180°$			
$0 < \theta < 90°$				$180° < \theta < 270°$			
$\theta = 90°$				$\theta = 270°$			
$90° < \theta < 180°$				$270° < \theta < 360°$			

Use the unit circle to determine each value in the table, using either “0”, “+ve”, “-ve”, “1”, “-1” or “undefined”. Recall that the point on the unit circle has coordinate $(\cos\theta, \sin\theta)$ and OP has gradient $\tan\theta$.

Gradient of OP.

Chapter 10 · Original slide 6 · The Unit Circle and Trigonometry



The unit circles explains the behaviour of these trigonometric graphs beyond $90°$.

However, the easiest way to remember whether $\sin(x), \cos(x), \tan(x)$ are positive or negative is to just do a very quick sketch (preferably mentally!) of the corresponding graph.

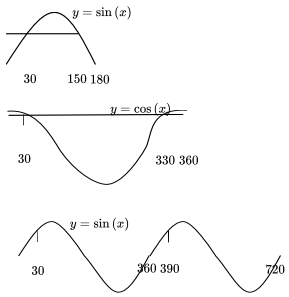
positive
for $0 < x < 180°$

negative
for $180° < x < 360°$

Note: The textbook uses something called ‘*CAST diagrams*’, which is an acronym for which quadrants the Cosine, All, Sine and Tangent are positive

Trigonometry

Chapter 10 · Original slide 7 · A Few Trigonometric Angle Laws



The following are all easily derivable using a quick sketch of a trigonometric graph, and are merely a convenience so you don't always have to draw out a graph every time.

You are highly encouraged to memorise these so that you can do exam questions faster.

1

$\sin(x) = \sin(180^\circ - x)$

e.g. $\sin(150^\circ) = \sin(30^\circ)$

e.g. $\cos(330^\circ) = \cos(30^\circ)$

e.g. $\sin(390^\circ) = \sin(30^\circ)$

Remember from the previous chapter that "cosine" by definition is the sine of the "complementary" angle.

This was/is never covered in the textbook but caught everyone by surprise when it came up in a C3 exam.

We saw this in the previous chapter when covering the 'ambiguous case' when using the sine rule.

2

$\cos(x) = \cos(360^\circ - x)$

3

sin and cos repeat every 360°

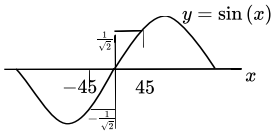
but tan every 180°

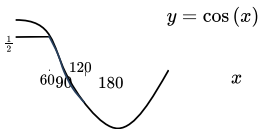
4

$\sin(x) = \cos(90^\circ - x)$

e.g. $\sin(50^\circ) = \cos(40^\circ)$

Chapter 10 · Original slide 8 · Examples





Without a calculator, work out the value of each below.

tan repeats every 180° so can subtract 180°

$\tan(225^\circ) =$

$\tan(210^\circ) =$

$\sin(150^\circ) =$

$\cos(300^\circ) =$

$\sin(-45^\circ) =$

Use the 'laws' where you can, but otherwise just draw out a quick sketch of the graph.

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every 360° but tan every 180°

For sin we can subtract from 180°.

Chapter 10 · Original slide 9 · Test Your Understanding

Without a calculator, work out the value of each below.

$\cos(315^\circ) =$

$\sin(420^\circ) =$

$\tan(-120^\circ) =$

$\tan(-45^\circ) =$

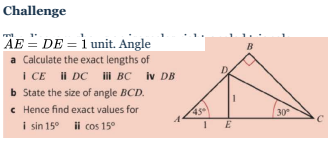
$\sin(135^\circ) =$

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- $\sin, \cos$ repeat every 360° but $\tan$ every 180°

Chapter 10 · Original slide 10 · Exercise 10B – Y1 Page 209

1 Express the following as trigonometric ratios of either 30°, 45° or 60°, *and hence find their exact values.* **a** $\sin 135^\circ$ **b** $\sin(-60^\circ)$ **c** $\sin 330^\circ$ **d** $\sin 420^\circ$ **e** $\sin(-300^\circ)$ **f** $\cos 120^\circ$ **g** $\cos 300^\circ$ **h** $\cos 225^\circ$ **i** $\cos(-210^\circ)$ **j** $\cos 495^\circ$ **k** $\tan 135^\circ$ **l** $\tan(-225^\circ)$ **m** $\tan 210^\circ$ **n** $\tan 300^\circ$ **o** $\tan(-120^\circ)$.

Chapter 10 · Original slide 11 · Exercise 10A/B – Y1 Page 207, 209



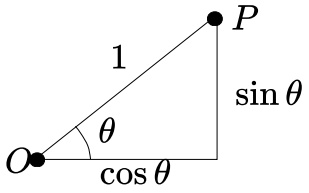
a Calculate the exact lengths of

i CE **ii** DC **iii** BC **iv** DB **b** State the size of angle BCD.

c Hence find exact values for

i $\sin 15°$ **ii** $\cos 15°$

Chapter 10 · Original slide 12 · Trigonometric Identities



Returning to our point on the unit circle...

1 2

Pythagoras gives you...

$\sin^2\theta$ is a shorthand for $(\sin\theta)^2$. It does NOT mean the sin is being squared – this does not make sense as sin is a function, and not a quantity that we can square!

Chapter 10 · Original slide 13 · Application of identities #1: Proofs

Prove that $1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

$\tan x = \frac{\sin(x)}{\cos(x)}$

$\sin^2 x + \cos^2 x = 1$

Tip #1: Turn any tan's into sin's and cos's.

Recall that $\equiv$ means 'equivalent to', and just means the LHS is always equal to the RHS for all values of θ.

From Chapter 7 ('Proofs') we saw that usually the best method is to manipulate one side (e.g. LHS) until we get to the other (RHS).

Chapter 10 · Original slide 14 · More Examples

Prove that $\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta}$

Tip #3: Look out for $1 - \sin^2\theta$ and $1 - \cos^2\theta$.

Students often don't spot that these can be simplified.

Simplify $5 - 5\sin^2\theta$

Tip #2: In any addition/subtraction involving at least one fraction (with trig functions), always combine algebraically into one.

Chapter 10 · Original slide 15 · Test Your Understanding

Prove that $\frac{\tan x \cos x}{\sqrt{1-\cos^2 x}} \equiv 1$

Prove that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

Prove that $\tan^2\theta \equiv \frac{1}{\cos^2\theta} - 1$

Chapter 10 · Original slide 16 · Exercise 10C – Y1 Page 211-212

10 In each of the following, eliminate θ to give an equation relating x and y:

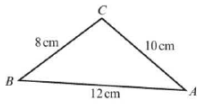
a $x = \sin\theta, y = \cos\theta$ **b** $x = \sin\theta, y = 2\cos\theta$

c $x = \sin\theta, y = \cos^2\theta$ **d** $x = \sin\theta, y = \tan\theta$

Problem-solving

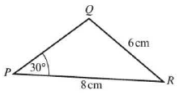
In part e find expressi

Chapter 10 · Original slide 17 · Exercise 10C – Y1 Page 211-212



11 The diagram shows the triangle ABC with $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.

a Show that $\cos B = \dfrac{9}{16}$. (3 ma



b Hence find the exact value of $\sin B$.

12 The diagram shows triangle PQR with $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and angle $QPR = 30°$.

a Show that $\sin Q = \dfrac{2}{3}$.

b Given that Q is obtuse, find the exact value of $\cos Q$.

Chapter 10 · Original slide 19 · Solving Trigonometric Equations

Remember those trigonometric angle laws (on the right) earlier this chapter? They're about to become very useful!

Reminder of 'trig laws':

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every $360°$ but tan every $180°$

Solve $\sin\theta = \frac{1}{2}$ in the interval $0 \leq \theta \leq 360°$.

Calculator Note:

When you do sin^{-1}, cos^{-1} and tan^{-1} on a calculator, it gives you only one value, known as the principal value.

Solve $5\tan\theta = 10$ in the interval $-180° \leq \theta < 180°$

Tip: Look out for the solution range required. $-180 \leq \theta < 180°$ is a particularly common one.

tan repeats every $180°$, so can add/subtract $180°$ as we please.

Chapter 10 · Original slide 20 · Slightly Harder Ones…

Solve $\sin\theta = -\frac{1}{2}$ in the interval $0 \leq \theta \leq 360^\circ$.

Note that we've had to use a second law, i.e. that $\sin$ repeats every 360°.

Hint: The problem here is that we have two different trig functions. Is there anything we can divide both sides by so we only have one trig function?

In general you should have 2 solutions per 360° (except when at a peak or trough of the trig graph)

Solve $\sin\theta = \sqrt{3}\cos\theta$ in the interval $0 \leq \theta \leq 360^\circ$.

Chapter 10 · Original slide 21 · Test Your Understanding

Solve $2\cos\theta = \sqrt{3}$ in the interval $0 \leq \theta \leq 360^\circ$.

Solve $\sqrt{3}\sin\theta = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.

Chapter 10 · Original slide 22 · Exercise 10D – Y1 215-216

7 A teacher asks two students to solve the equation $2\cos x = 3\sin x$ for $-180^\circ \leq x \leq 180^\circ$. The attempts are shown:
Student A: $\tan x = \frac{3}{2}$, $x = 56.3^\circ$ or $x = -123.7^\circ$.
Student B: $4\cos^2 x = 9\sin^2 x$,
$4(1 - \sin^2 x) = 9\sin^2 x$, $4 = 13\sin^2 x$,
$\sin x = \pm\sqrt{\frac{4}{13}}$, $x = \pm 33.7^\circ$ or $x = \pm 146.3^\circ$.
a Identify the mistake made by Student A. (1 mark)
b Identify the mistake made by Student B and explain the effect it has on their solution. (2 marks)
c Write down the correct answers to the question. (1 mark)

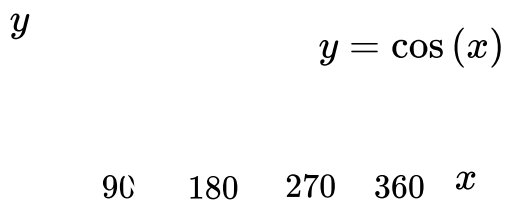
9 Find all the values of θ, to 1 decimal place, in the interval $0 < \theta < 360^\circ$ for which $\tan^2\theta = 9$

Chapter 10 · Original slide 23 · Exercise 10D – Y1 215-216

10 a Show that $4\sin^2 x - 3\cos^2 x = 2$ can be written as $7\sin^2 x = 5$.
b Hence solve, for $0 \leq x \leq 360^\circ$, the equation $4\sin^2 x - 3\cos^2 x = 2$. Give your answers to 1 decimal place.

11 a Show that the equation $2\sin^2 x + 5\cos^2 x = 1$ can be written as $3\sin^2 x = 4$.
b Use your result in part a to explain why the equation $2\sin^2 x + 5\cos^2 x = 1$ has no solutions.

Chapter 10 · Original slide 24 · Harder Equations



Harder questions replace the angle θ with a linear expression.

Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360°$.

STEP 1: Adjust the range of values for θ to match the expression inside the cos.

STEP 2: Immediately after applying an inverse trig function (and BEFORE dividing by 3!), find all solutions up to the end of the interval.

In general you tend to get a pair of values per $360°$ (for any of sin/cos/tan), except for $\cos\theta = \pm 1$ or $\sin\theta = \pm 1$:

Thus once getting your first pair of values (e.g. using $\sin(180 - \theta)$ or $\cos(360 - \theta)$ to get the second value), keep adding $360°$ to generate new pairs.

STEP 3: Then do final manipulation to each value.

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Solve $\sin(2x + 30°) = \frac{1}{\sqrt{2}}$ in the interval $0 \leq x \leq 360°$.

To get from x to $2x + 30°$ we double and add $30°$. So do the same to the upper and lower bound!

Solve $\sin x = 2\cos x$ in the interval $0 \leq x < 300°$

By dividing both sides by $\cos x$, the $\sin x$ becomes $\tan x$ and the $\cos x$ disappears, leaving a trig equation helpfully only in terms of one trig function.

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Solve, for $0 \leq x < 180^\circ$,
$\cos(3x - 10^\circ) = -0.4$,
giving your answers to 1 decimal place. You should show each step in your working. (7)

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4 Solve for $0 \leq x \leq 180^\circ$ the equations:
a $\sin(x + 20^\circ) = \frac{1}{2}$.
b $\cos 2x = -0.8$, giving your answers to 1 decimal place.

5 a Sketch for $0 \leq x \leq 360^\circ$ the graph of $y = \sin(x + 60^\circ)$.
b Write down the exact coordinates of the points where the graph meets the coordinate axes.
c Solve, for $0 \leq x \leq 360^\circ$, the equation $\sin(x + 60^\circ) = 0.55$, giving your answers to 1 decimal place.

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6 a Given that $4\sin x = 3\cos x$, write down the value of $\tan x$.
b Solve, for $0 \leq \theta \leq 360°$, $4\sin 2\theta = 3\cos 2\theta$ giving your answers to 1 decimal place.

7 The equation $\tan kx = -\frac{1}{\sqrt{3}}$, where k is a constant and $k > 0$, has a solution at $x = 60°$.
a Find a possible value of k.
b State, with justification, whether this is the only such possible value of k.

Chapter 10 · Original slide 29 · Quadratics in sin/cos/tan

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Method 1: Use a substitution.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360°$.

Method 2: Factorise without substitution.

This is the same, but we 'imagine' $\sin x$ as a single variable and hence factorise immediately.

Chapter 10 · Original slide 30 · More Examples

Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360°$.

Missing the negative case would result in the loss of multiple marks. Beware!

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180° \leq x \leq 180°$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

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(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360°$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

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5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

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7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

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9 $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540^\circ$.